

SDS-ATP · ROOM ALLOCATION ILP

Mathematical Formulation

Full Integer Linear Program for `utils/gurobi_solver_grouped.py` (`allot_rooms`): sets, variables, constraints, and lexicographic objectives.

1. Sets and indices

Symbol	Meaning
$E = \{1, \dots, n_e\}$	Student exam-rows (one row = one student sitting one exam)
$R = \{1, \dots, n_r\}$	Room availability slots (rows with positive testing capacity)
C	Distinct course IDs appearing in E
G	Exam cohorts : students with the same (Course_ID, Date, Time_Start, Time_End)
$E_g \subseteq E$	Student rows belonging to cohort $g \in G$
$C_c \subseteq E$	Student rows with course $c \in C$
$E^{\text{RD}} \subseteq E$	Rows tagged <code>RD</code>
$E^{\text{PRIV}} \subseteq E$	Rows tagged <code>PRIV</code> or <code>CODS</code>

Compatibility and time structure

Symbol	Meaning
$\mathcal{A} \subseteq E \times R$	Compatible pairs (i, j) : same date, exam fits in room window, and room opens ≥ 15 min before exam start
$R_i = \{j : (i, j) \in \mathcal{A}\}$	Rooms compatible with exam i
$E_j = \{i : (i, j) \in \mathcal{A}\}$	Exams compatible with room j
$R_g = \bigcup_{i \in E_g} R_i$	Rooms compatible with at least one member of cohort g
T_j	Distinct exam start times (minutes) among exams in E_j
$E_{j,t} \subseteq E_j$	Concurrent group: exams in slot j with start time t
$\mathcal{B}_{30} \subseteq \mathcal{A}$	Pairs with ≥ 30 min pre-exam open buffer
$\mathcal{B}_{15}^{\text{post}} \subseteq \mathcal{A}$	Pairs with ≥ 15 min post-exam buffer before room close
$\mathcal{A}_{15} = \mathcal{A} \setminus \mathcal{B}_{30}$	Compatible pairs with only a 15–29 min pre-buffer
$\mathcal{A}_0^{\text{post}} = \mathcal{A} \setminus \mathcal{B}_{15}^{\text{post}}$	Compatible pairs lacking a 15 min post-buffer
$R^{\text{Z2}} \subseteq R$	Room slots with Zone $\neq 1$
$\mathcal{F}_{15} \subseteq E \times E \times R$	Non-concurrent pairs in slot j with inter-exam gap < 15 min (hard conflict)
$\mathcal{F}_{15-29} \subseteq E \times E \times R$	Non-concurrent pairs in slot j with gap in $[15, 30)$ min (soft penalty)

For non-concurrent i_1, i_2 in the same slot j , the gap is

$$\text{gap}(i_1, i_2) = \max(t_{i_2}^{\text{start}} - t_{i_1}^{\text{end}}, t_{i_1}^{\text{start}} - t_{i_2}^{\text{end}}).$$

Same start time $\Rightarrow$ concurrent (no gap rule).

Room and exam parameters

Symbol	Meaning
$\text{cap}_j \in \mathbb{Z}_{>0}$	Testing capacity of room slot j
$t_i^{\text{start}}, t_i^{\text{end}}$	Exam start/end (minutes since midnight)
$r_j^{\text{open}}, r_j^{\text{close}}$	Room slot open/close (minutes)

Compatibility (C6, enforced by restricting $\mathcal{A}$):

$$(i, j) \in \mathcal{A} \iff \text{date}_i = \text{date}_j \wedge t_i^{\text{start}} - 15 \geq r_j^{\text{open}} \wedge t_i^{\text{end}} \leq r_j^{\text{close}}.$$

2. Decision variables

All variables are binary.

Variable	Domain	Meaning
x_{ij}	$(i, j) \in \mathcal{A}$	1 if exam-row i is assigned to room slot j
y_j	$j \in R$	1 if room slot j is used by any assigned exam
u_{gj}	$g \in G, j \in R_g$	1 if cohort g places at least one student in room j
z_{cjt}	$c \in C, j \in R, t \in T_j$ (when course c appears among candidates in $E_{j,t}$)	1 if course c has ≥ 1 assigned student in concurrent group (j, t)
ρ_{jt}	$j \in R, t \in T_j$	1 if concurrent group (j, t) contains an assigned RD student
$q_{i_1 i_2 j}$	$(i_1, i_2, j) \in \mathcal{F}_{15-29}$	1 if both i_1 and i_2 are assigned to j with only a 15–29 min gap

$$x_{ij}, y_j, u_{gj}, z_{cjt}, \rho_{jt}, q_{i_1 i_2 j} \in \{0, 1\}.$$

3. Constraints

C1 — At most one room per exam

$$\sum_{j \in R_i} x_{ij} \leq 1 \quad \forall i \in E \text{ with } R_i \neq \emptyset.$$

Link y — Room usage

$$x_{ij} \leq y_j \quad \forall (i, j) \in \mathcal{A}.$$

Link u — Cohort–room indicator

For each $g \in G$, $j \in R_g$, let $E_g(j) = \{i \in E_g : (i, j) \in \mathcal{A}\}$.

$$\begin{aligned} u_{gj} &\geq x_{ij} && \forall i \in E_g(j), \\ u_{gj} &\leq \sum_{i \in E_g(j)} x_{ij} && (\text{if } E_g(j) \neq \emptyset). \end{aligned}$$

So $u_{gj} = 1$ iff at least one member of cohort g is assigned to j .

C2 — Capacity per concurrent group

$$\sum_{i \in E_{j,t}} x_{ij} \leq \text{cap}_j \quad \forall j \in R, t \in T_j.$$

C3 — At most three courses per concurrent group

Linking for course indicators (standard big- M / indicator bounds):

$$\begin{aligned} z_{cjt} &\geq x_{ij} && \forall i \in E_{j,t} \cap C_c, \\ z_{cjt} &\leq \sum_{i \in E_{j,t} \cap C_c} x_{ij} && (\text{when } E_{j,t} \cap C_c \neq \emptyset), \\ \sum_{\substack{c \in C \\ z_{cjt} \text{ defined}}} z_{cjt} &\leq 3 && \forall j \in R, t \in T_j. \end{aligned}$$

C4 — RD tag $\Rightarrow$ concurrent group capped at 20

If $\text{cap}_j > 20$ and $E_{j,t} \neq \emptyset$:

$$\begin{aligned} \rho_{jt} &\geq x_{ij} && \forall i \in E_{j,t} \cap E^{\text{RD}}, \\ \sum_{i \in E_{j,t}} x_{ij} &\leq \text{cap}_j - (\text{cap}_j - 20) \rho_{jt}. \end{aligned}$$

When $\rho_{jt} = 0$, the right-hand side is cap_j ; when $\rho_{jt} = 1$, it becomes 20.

C5 — PRIV / CODS alone in concurrent group

For each $i \in E^{\text{PRIV}}$, let $t_i = t_i^{\text{start}}$ and $E_{j,t_i}^{(-i)} = E_{j,t_i} \setminus \{i\}$.

$$\sum_{k \in E_{j,t_i}^{(-i)}} x_{kj} \leq \text{cap}_j (1 - x_{ij}) \quad \forall j \in R_i \text{ with } E_{j,t_i}^{(-i)} \neq \emptyset.$$

If $x_{ij} = 1$, no other exam may share concurrent group (j, t_i) .

C7 — Inter-exam spacing (hard)

$$x_{i_1 j} + x_{i_2 j} \leq 1 \quad \forall (i_1, i_2, j) \in \mathcal{F}_{15}.$$

Soft-gap linking for q (used in P2)

$$q_{i_1 i_2 j} \geq x_{i_1 j} + x_{i_2 j} - 1 \quad \forall (i_1, i_2, j) \in \mathcal{F}_{15-29}.$$

(Minimizing $\sum q$ in P2 forces $q = 1$ only when both exams are assigned with a tight 15–29 min gap.)

4. Multi-objective (lexicographic)

Gurobi `setObjectiveN` with **minimize** sense. Higher priority is optimized first; later priorities are optimized without worsening earlier ones.

(P5)	$\min - \sum_{(i,j) \in \mathcal{A}} x_{ij}$	(maximize assignments)
(P4)	$\min \sum_{g \in G} \sum_{j \in R_g} u_{gj}$	(minimize rooms per cohort / avoid splits)
(P3)	$\min \sum_{j \in R} y_j$	(minimize total room slots used)
(P2)	$\min \sum_{(i,j) \in \mathcal{A}_{15}} x_{ij} + \sum_{(i_1, i_2, j) \in \mathcal{F}_{15-29}} q_{i_1 i_2 j}$	(prefer 30 min pre-/inter-exam buffers)
(P1)	$\min \sum_{(i,j) \in \mathcal{A}_0^{\text{post}}} x_{ij}$	(prefer 15 min post-exam buffer)
(P0)	$\min \sum_{\substack{(i,j) \in \mathcal{A} \\ j \in R^{Z2}}} x_{ij}$	(prefer Zone 1 over Zone 2).

Priority order in code: P5 > P4 > P3 > P2 > P1 > P0 (priorities 5, 4, 3, 2, 1, 0).

5. Compact statement of the model

$$\begin{aligned}
& \text{lex-min} \quad \left(-\sum_{(i,j) \in \mathcal{A}} x_{ij}, \sum_{g,j} u_{gj}, \sum_j y_j, \sum_{(i,j) \in \mathcal{A}_{15}} x_{ij} + \sum_{(i_1,i_2,j) \in \mathcal{F}_{15-29}} q_{i_1 i_2 j}, \sum_{(i,j) \in \mathcal{A}_0^{\text{post}}} x_{ij}, \sum_{(i,j)} \right. \\
& \text{s.t.} \quad \sum_{j \in R_i} x_{ij} \leq 1 \\
& \quad x_{ij} \leq y_j \\
& \quad u_{gj} \geq x_{ij}, \quad u_{gj} \leq \sum_{i \in E_g(j)} x_{ij} \\
& \quad \sum_{i \in E_{j,t}} x_{ij} \leq \text{cap}_j \\
& \quad z_{cjt} \geq x_{ij} \ (i \in E_{j,t} \cap C_c), \quad z_{cjt} \leq \sum_{i \in E_{j,t} \cap C_c} x_{ij}, \quad \sum_c z_{cjt} \leq 3 \\
& \quad \rho_{jt} \geq x_{ij} \ (i \in E_{j,t} \cap E^{\text{RD}}), \\
& \quad \sum_{i \in E_{j,t}} x_{ij} \leq \text{cap}_j - (\text{cap}_j - 20)\rho_{jt} \\
& \quad \sum_{k \in E_{j,t_i}^{(-i)}} x_{kj} \leq \text{cap}_j(1 - x_{ij}) \\
& \quad x_{i_1 j} + x_{i_2 j} \leq 1 \\
& \quad q_{i_1 i_2 j} \geq x_{i_1 j} + x_{i_2 j} - 1 \\
& \quad x, y, u, z, \rho, q \in \{0, 1\}.
\end{aligned}$$

6. Post-solve booking windows (not part of the ILP)

Exam row times $t_i^{\text{start}}, t_i^{\text{end}}$ are **not** modified. For each assigned pair with $x_{ij}^* = 1$, a booking window is recorded:

$$\text{pre}_{ij} = \begin{cases} 30 & \text{if } (i, j) \in \mathcal{B}_{30}, \\ 15 & \text{otherwise,} \end{cases} \quad \text{post}_{ij} = \begin{cases} 15 & \text{if } (i, j) \in \mathcal{B}_{15}^{\text{post}}, \\ 0 & \text{otherwise,} \end{cases}$$

$$\text{BookStart}_{ij} = t_i^{\text{start}} - \text{pre}_{ij}, \quad \text{BookEnd}_{ij} = t_i^{\text{end}} + \text{post}_{ij}.$$

The returned `bookings` table is the set of distinct (Date, Room, BookStart, BookEnd) tuples.

7. Variable / constraint map to code

Math	Code (gurobi_solver_grouped.py)
x_{ij}	<code>x[(i, j)]</code>
y_j	<code>y[j]</code>
u_{gj}	<code>u[(g, j)]</code>
z_{cjt}	<code>z_g[(c, j, ts)]</code>
ρ_{jt}	<code>rd_flag_g[(j, ts)]</code>
$q_{i_1 i_2 j}$	<code>q[(i1, i2, j)]</code>
$\mathcal{A}$	<code>compat</code>
$\mathcal{B}_{30}$	<code>has_30_buffer</code>
$\mathcal{B}_{15}^{\text{post}}$	<code>has_15_post_buffer</code>
$\mathcal{F}_{15}$	<code>inter_15_blocked</code>
$\mathcal{F}_{15-29}$	<code>inter_only_15</code>
$E_{j,t}$	<code>groups[j][ts]</code>
G, E_g	<code>group_keys , exam_groups</code>